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CRFusion: A Novel LiDAR-Camera Fusion Network for BEV Map Construction

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Abstract

The generation of high-definition semantic maps for the environment is a crucial component in automatic driving. Many methods based on different sensor fusion to achieve this key goal are proposed, and good performances have been obtained such as LiDAR and cameras. However, the issue of missing information in the part of maps still persists. To resolve the above problem, a novel LiDAR-Camera fusion network for Bird's Eye View (BEV) map construction, **CRFusion**, is proposed. Firstly, the Long-Range LiDAR Feature Prediction (LRP) method is introduced, in which cross-attention is employed to enable interaction between image BEV features and LiDAR BEV features, and a self-attention mechanism is utilized to enhance the LiDAR BEV features. Secondly, to overcome the limitations of traditional fusion strategies on different lighting and weather conditions, the Feature Alignment and Dynamic Gated Fusion (DGF) methods are designed, which allow for spatial alignment of LiDAR and camera features. Fusion weights based on environmental conditions are adaptively adjusted, thereby the stability of the fusion effect is ensured. Finally, the feasibility of CRFusion is validated on the Mini NuScenes dataset and the experiment results conducted on the NuScenes public dataset demonstrate that the CRFusion method outperforms other methods. The superiority of the proposed method is thus confirmed.

1.Introduction

The detection of street lanes and the generation of semantic maps are crucial for autonomous vehicles to achieve full autonomy¹⁻⁴. Recently, semantic map construction and local map completion have become key components for enabling autonomous navigation and environmental understanding in autonomous vehicles⁵. Semantic map construction^{6,7} is formulated as a semantic segmentation task in the BEV space, where each pixel in the BEV plane is assigned a semantic label. Local map completion, in contrast, focuses on inferring occluded or missing map structures by leveraging the complementary strengths of cameras and LiDAR, particularly when a single sensor modality is limited. In response to these task requirements, map construction methods can be categorized based on the type of sensor input: camera-based methods⁸⁻¹¹, LiDAR-based methods^{12,13}, and camera-LiDAR fusion methods¹⁴⁻¹⁹.

In the field of LiDAR-based semantic map construction, early studies primarily focused on point cloud segmentation²⁰. For example, RangeNet++²¹ achieves rapid segmentation by projecting the point

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cloud into a range image, albeit at the cost of a loss in geometric detail. Cylinder3D²² employs a cylindrical partition strategy and asymmetric convolution blocks, thereby enhancing the segmentation accuracy for sparse point clouds. PointPillars¹² generates a pseudo-image from a pillar-based representation, paving the way for BEV map construction. Subsequently, PolarNet²³ leverages a polar coordinate system for BEV grid discretization, making it better suited for the requirements of surround-view perception. Despite providing precise 3D geometric information on the LiDAR, LiDAR-based methods often suffer from semantic information loss at medium-to-long ranges due to the inherent sparsity of point clouds. In contrast, camera-based approaches hold a pronounced advantage in semantic acquisition at these distances.

In constructing semantic maps based on stereo vision, LMSC²⁴ as one of early approaches estimates depth using stereo disparity and predicts voxel semantics via a 3D convolution network, which suffers from high computational cost and sensitivity to depth errors. HDNet²⁵ achieves high-resolution map reconstruction by implicitly representing scene geometry and semantics. But there is the limitation of its adaptability to complex dynamic scenes. In recent years, BEVFormer²⁶ leverages a Transformer architecture to learn an end-to-end mapping from multi-view images to a BEV representation, thereby avoiding the need for explicit depth estimation and enhancing the robustness of image features. M²BEV²⁷ further unifies the tasks of map segmentation and object detection within a unified BEV space. However, camera-based methods for semantic mapping generally lack precise depth information. To achieve high accuracy, high-resolution input images or larger backbone networks are relied on.

Recently, methods based on the LiDAR and camera have significantly improved the reliability and completeness of environmental perception by combining the geometric information from LiDAR data with the rich semantics from camera data. PointPainting²⁸ as one of the preliminary methods achieves feature-level fusion by projecting image semantics onto the LiDAR point cloud. But the critical issue of cross-modal feature alignment is not adequately addressed. MVP¹⁹ enhances cross-modal representation capability by projecting image features into the 3D space. In recent years, BEVFusion²⁹ significantly advances the field by achieving efficient fusion in a unified BEV space, balancing accuracy and speed. Superfusion³⁰ utilizes attention mechanisms to inject image features into sparse BEV representations, extending semantic map coverage to 90 meters. However, the above methods suffer performance limitations in the extreme weather or lighting conditions, and static fusion strategies struggle to adapt to dynamic scene changes. Therefore, there is an urgent need to develop a robust method of LiDAR-camera fusion semantic map generation method to support autonomous driving applications in complex urban environments.

The paper focuses on semantic segmentation techniques in complex urban environments and the CRFusion algorithm framework based on multimodal fusion is proposed. The primary contributions of this study are as follows:

1. To address the issue of BEV feature fusion in map construction tasks, we propose CRFusion, which leverages complementary BEV features from different sensors.

2. To address the issue of missing local or distant features in semantic maps generated by fusing LiDAR features with forward-looking image features, we propose a LiDAR BEV feature prediction method based on a LiDAR Range Predictor (LRP). This approach facilitates interaction between the two BEV feature spaces and enhances feature representation through a self-attention mechanism.

3. For better feature fusion, we propose an effective Dynamic Gated Fusion (DGF) module to adaptively select valuable information from different modalities.

2. Related works

Our work is highly related to map construction task and multi-sensor fusion methods, which will be discussed thoroughly in the following.

2.1 BEV map segmentation

Bird's Eye View map segmentation serves as the core paradigm for autonomous driving perception. Its fundamental value lies in providing a unified, perspective-free spatial representation, acting as an ideal interface for multimodal fusion and temporal information alignment. BEVFormer²⁶ implicitly constructs BEV features by leveraging learnable queries and spatial temporal attention mechanisms. Current research focuses have shifted toward multimodal BEV fusion (e.g., BEVFusion²⁹), temporal-spatial deep integration, and evolution toward 3D occupancy networks. The latter aims to describe occupancy and semantic information across the entire 3D space.

2.2 Multi-sensor fusion

Multi-sensor fusion is a core technology for enhancing the robustness and accuracy of autonomous driving systems. Its evolution has progressed from early decision-level post-fusion to the current mainstream feature-level deep fusion, where Bird's eye view space has become the standard interface for achieving multi-modal feature alignment and fusion. Building upon this foundation, symmetric fusion paradigms like BEVFusion²⁹ achieve significant performance gains by adaptively integrating camera and LiDAR features within a unified BEV space. Meanwhile, approaches such as TransFusion³⁶ employ query mechanisms to enable refined fusion between LiDAR initial proposals and image semantic features. Further enhanced the robustness of autonomous³⁷⁻³⁹ driving target detection.

2.3 Comparison with Existing Works

This work differs from previous literature in three key aspects. Firstly, we focus on the BEV based map construction task, which is a semantically oriented task that places greater emphasis on semantic information within images. Secondly, addressing the issue of missing local or long-range features in semantic maps generated by fusing LiDAR and forward-view image features, we propose a LiDAR BEV feature prediction method based on a LiDAR Range Predictor. It enables interaction between the two BEV feature spaces and enhances feature representations through a self-attention mechanism. Furthermore, to address semantic misalignment between camera and LiDAR BEV features, we designed an AlignNet based BEV feature alignment method to improve feature registration. Finally, to further fuse multi-modal features, we propose an effective Dynamic Gated Fusion³¹⁻³³ method that adaptively selects valuable information from different modalities.

3. Methodology

To address the issues of semantic map generation gaps and incomplete local map completion in urban driving environments, a multimodal fusion map construction method, CRFusion is proposed. Firstly, the point cloud and image data are preprocessed to extract the BEV features of LiDAR and camera, respectively. Secondly, to enhance the long-range BEV features of LiDAR, the LiDAR Range Predictor(LRP) method is designed to improve the prediction capabilities of distant LiDAR features through image data. Next, to further improve the alignment of BEV features between LiDAR and

camera, the AlignNet method is introduced, which generates camera BEV features based on LiDAR BEV features. Finally, to enhance the fusion performance of LiDAR BEV features and camera BEV features in urban driving environments, the DGF method is proposed. The fused BEV features are input into the decoder and prediction head, ultimately outputting a complete semantic map. The specific structure is shown in Fig. 1.

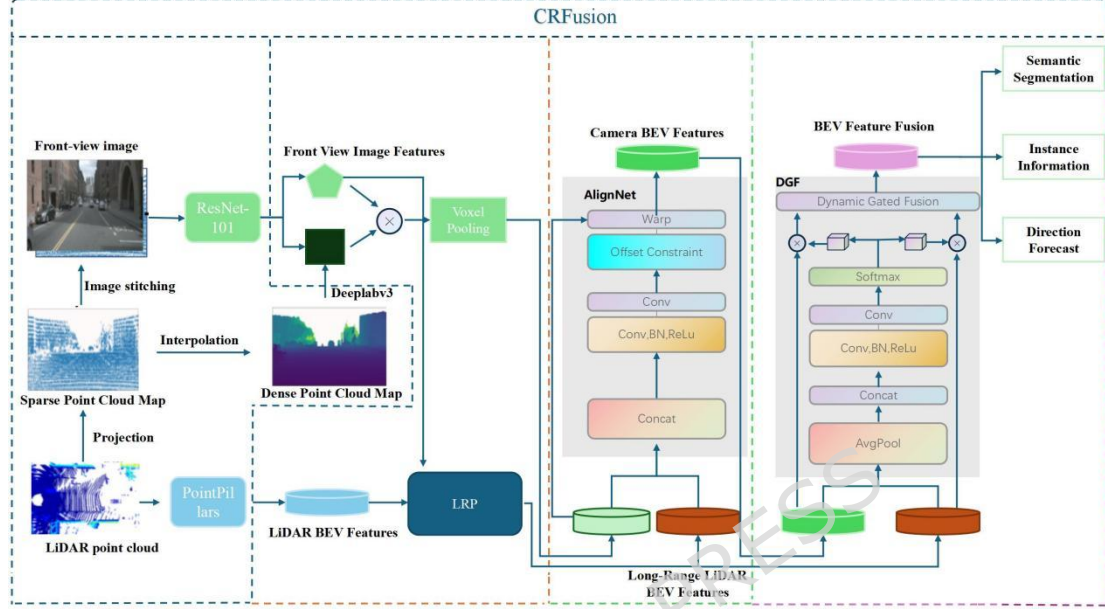


Fig. 1 Overall Structure of the CRFusion Algorithm

3.1 Data Preprocessing for LiDAR and Camera Systems

LiDAR signature extraction

Since LiDAR point clouds consist of a set of unordered, irregular, and sparsely distributed points in 3D space, the PointPillars method is adopted, which employs a pillarization preprocessing step to efficiently convert the data into a structured BEV representation. This process preserves essential spatial information while generating the initial LiDAR BEV features.

To mitigate the sparsity of LiDAR features at long range, a sparse depth map is generated at the raw data level. Firstly, the point cloud is transformed from the LiDAR coordinate system to the camera coordinate system using the rigid body transformation extrinsic parameters (rotation matrix R and translation vector T). Then, it is projected onto the image plane via the camera intrinsic matrix to obtain pixel coordinates (u, v) . Finally, the depth value (z -coordinate) from the camera frame is assigned to the corresponding pixel.

Image feature extraction

To balance deep semantic understanding with the preservation of spatial details, we construct the input by fusing a sparse depth map with a front-view image. A ResNet-101 backbone network is employed to extract high-level semantic features. Furthermore, a dense depth map is generated from the sparse one via depth discretization and a Deeplabv3 based depth prediction network. The accuracy of depth estimation is enhanced through the supervision by a one-hot encoded vector. This approach improves the model's robustness to scale-varying objects and provides a reliable data foundation for predicting long-range LiDAR features.

3.2 Long-Range LiDAR Feature Prediction Based on the LiDAR Range

Predictor Method

To address the issue of local or long-range feature gaps in semantic maps generated by fusing LiDAR features with forward-looking image features, we propose a prediction method of LiDAR BEV feature based on a LiDAR Range Predictor(LRP). During the long-range feature prediction process, a cross-attention mechanism is introduced to facilitate the interaction between the front-view image features and LiDAR BEV features, thereby compensating for the missing semantic information in the LiDAR void regions. Furthermore, a self-attention mechanism is employed to strengthen the contextual representation capability of the LiDAR BEV features, thereby enhancing the completeness of the feature in long-range scenarios. The specific structure is shown in Fig. 2.

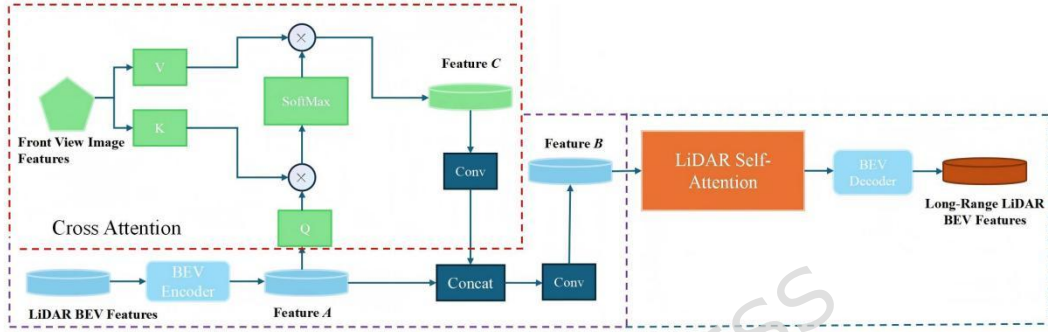


Fig. 2 Structure of the LiDAR Range Predictor Method

Long-Range LiDAR Feature Prediction Based on Cross-Attention

Compared with LiDAR, the camera is generally capable of perceiving ground areas at longer distances. Consequently, we propose an image-guided BEV prediction method that infers ground information within LiDAR blind spots by leveraging image features. The approach employs a cross-attention mechanism to capture correlation between LiDAR features and forward-looking image features, thereby achieving cross-modal information complementarity.

When long-range LiDAR features are predicted, the total number of elements along the spatial dimensions of the LiDAR feature tensor remains constant. Therefore, the tensor is reshaped to flatten its spatial dimensions and transposed into a standard sequential format to meet the input requirements of the attention mechanism. The processed LiDAR features are applied in the query vector $\mathbf{Q}$ in the attention module, while the image features serve as the key $\mathbf{K}$ and value $\mathbf{V}$ vectors, after the same flattening and transposing operations are undergone. The attention affinity matrix is computed by calculating the inner product between the query vector $\mathbf{Q}$ and the key vector $\mathbf{K}$, followed by normalization through the softmax operator. The normalized matrix is used to perform weighted aggregation of the value vectors $\mathbf{V}$, producing the feature representation $\mathbf{C}$, as shown in Equation (1). Subsequently, the aggregated feature $\mathbf{C}$ is concatenated with the original LiDAR feature $\mathbf{A}$ to form the fused feature matrix $\mathbf{F}$, as expressed in Equation (2).

$$\mathbf{C} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (1)$$

$$\mathbf{F} = [\mathbf{L} \oplus \mathbf{C}] = \begin{bmatrix} l_{1,1}^1 & \cdots & l_{1,1}^{256} & a_{1,1}^1 & \cdots & a_{1,1}^{256} \\ \vdots & & \vdots & \vdots & & \vdots \\ l_{H,W}^1 & \cdots & l_{H,W}^{256} & a_{H,W}^1 & \cdots & a_{H,W}^{256} \end{bmatrix} \quad (2)$$

where d_k represents the channel dimension used for scaling. To reduce the number of channels, the

fused feature matrix F undergoes convolution processing to derive the image-feature-guided long-range LiDAR feature B , yielding the output matrix G .

Local Enhancement of LiDAR BEV Features Based on Self-Attention Mechanism

To effectively capture long-range spatial dependencies and highlight important region features, a self-attention mechanism for the long-range LiDAR features B is introduced to enhance local features and strengthen their contextual awareness. The self-attention mechanism allows each element in the sequence to engage in a global interaction by generating a query Q' that represents its own needs, comparing with the keys K' of all elements in the sequence (including itself). The relevance is evaluated, the value V' information of all elements (including itself) is aggregated based on the evaluation results (i.e., attention weights). The final output is a new representation that integrates global contextual relationships and is rich in semantic associations, as presented in Fig. 3.

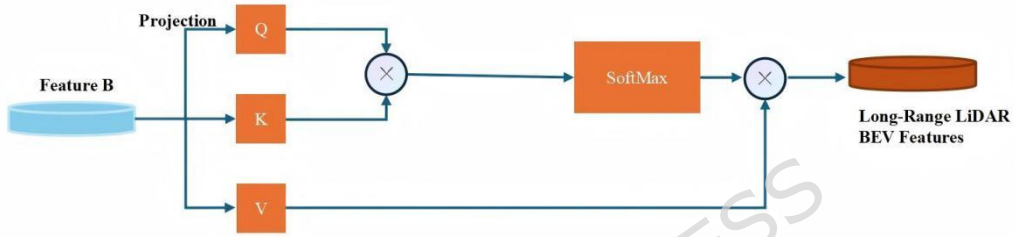


Fig. 3 Self-Attention Mechanism Architecture

Firstly, a linear projection of the enhanced feature B to generate the query Q' , key K' and value V' vectors is performed, as detailed in Equation (3).

$$\begin{cases} Q' = G \square W_q & W_q \square \mathbb{R}^{d_{\text{model}} \times d_k} \\ K' = G \square W_k & W_k \square \mathbb{R}^{d_{\text{model}} \times d_k} \\ V' = G \square W_v & W_v \square \mathbb{R}^{d_{\text{model}} \times d_v} \end{cases} \quad (3)$$

where G represents the sequence matrix for feature B , and d_{model} represents the embedding dimension. Subsequently, the attention affinity matrix is computed via the inner product between Q' and K' , to prevent vanishing gradients caused by excessively large dot product results, the attention scores are scaled as shown in Equation (4).

$$Scores_{ij} = \frac{Q' \cdot K'}{\sqrt{d_k}} \quad (4)$$

where $Scores_{ij}$ represents the score of position i relative to position j .

Subsequently, the scaled score matrix is normalized by the softmax operator, as shown in Equation (5).

$$A = \text{softmax}(Scores_{\text{scaled}}) \square \mathbb{R}^{n \times n} \quad (5)$$

By computing the weighted aggregated value V' as defined in Equation (6), the final LiDAR BEV feature Z is obtained.

$$Z = A \cdot V' \square \mathbb{R}^{n \times d_v} \quad (6)$$

3.3 AlignNet Method: Aligning Lidar Features with Camera Features

Due to the errors in dense point cloud depth estimation and inaccuracies in the external parameters in the projection of LiDAR point cloud, there is the misalignment between the BEV features from the camera and LiDAR branches. The fusion performance is degraded when LiDAR BEV features are directly connected with camera BEV features. To address the issue, a BEV feature alignment method based on AlignNet is designed to improve feature alignment, as depicted in Fig 4.

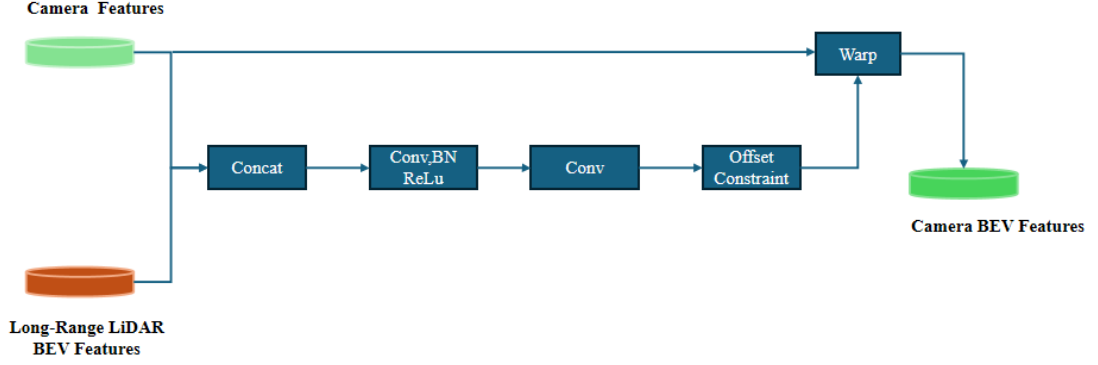


Fig. 4 Structure of the AlignNet Method

The camera BEV features and the enhanced LiDAR BEV features $\mathbf{Z}$ are concatenated along the channel dimension as input. A lightweight convolution network is used to predict the spatial offset Δ for each pixel, as described in Equation (7). And the predicted offset is applied to deform the original grid coordinates. The camera BEV features undergo differentiable resampling on a bilinear interpolation mechanism, generating corrected features which are aligned with the LiDAR BEV feature space. The alignment mechanism effectively mitigates the feature map misalignment caused by depth estimation errors and inaccuracies in the external parameters used for point cloud projection, as shown in Equation (8). The overall structure enables end-to-end feature alignment, significantly enhancing the robustness and consistency of multimodal BEV feature fusion.

$$\Delta = f_{\Delta}(\text{concat}) = \text{Conv}_{3 \times 3} \left(\sigma \left(\text{BN}(\text{Conv}_{1 \times 1}(\mathbf{Z} \square \mathbf{C})) \right) \right) \quad (7)$$

$$\mathbf{C}''_{wh} = \sum_{w'=1}^W \sum_{h'=1}^H \mathbf{C}'_{w'h'} \square \max(0, 1 - |w + \Delta_{1wh} - w'|) \square \max(0, 1 - |h + \Delta_{2wh} - h'|) \quad (8)$$

where $\mathbf{C}''_{wh}$ represents the value of the camera's BEV feature after alignment at the position, and $\sum_{w'=1}^W \sum_{h'=1}^H$ represents the double summation over all possible image line positions w', h' , which means considering the contribution of each pixel in the original camera BEV feature $\mathbf{C}$. $\mathbf{C}'_{w'h'}$ represents the feature value at position (w', h') in the original camera BEV feature, while $\max(0, 1 - |w + \Delta_{1wh} - w'|)$ and $\max(0, 1 - |h + \Delta_{2wh} - h'|)$ correspond to the bilinear interpolation weights, which compute the horizontal and vertical offsets of the original feature position w', h' relative to the target position (w, h) . The weight decreases with increasing distance after the flow field offset. If the distance exceeds a certain range, the contribution is set to zero. $\Delta_{w_{wh}}$ and $\Delta_{h_{wh}}$ represent the flow field values at position (w, h) , respectively, indicating the horizontal and vertical offsets from the original position to the target position.

3.4 Dynamic Gated Fusion Method: Adaptive Integration of Lidar and Visual Features

To address the issue where LiDAR BEV features and camera BEV features directly concatenated lead to a decline in fusion performance in different weather conditions (daytime and nighttime) or

urban environments, a feature fusion method based on Dynamic Gated Fusion (DGF) is proposed. Compared to methods that rely on precise geometric estimation, it demonstrates superior robustness relative to complex global attention mechanisms, and it offers greater efficiency and a more focused objective. In contrast to static fusion approaches, it possesses the crucial capability of scene adaptation. The method adaptively evaluates and fuses the weights and relevant information of the two modalities, further enhancing the multimodal feature fusion performance at the BEV level. The specific structure is illustrated in Fig. 5.

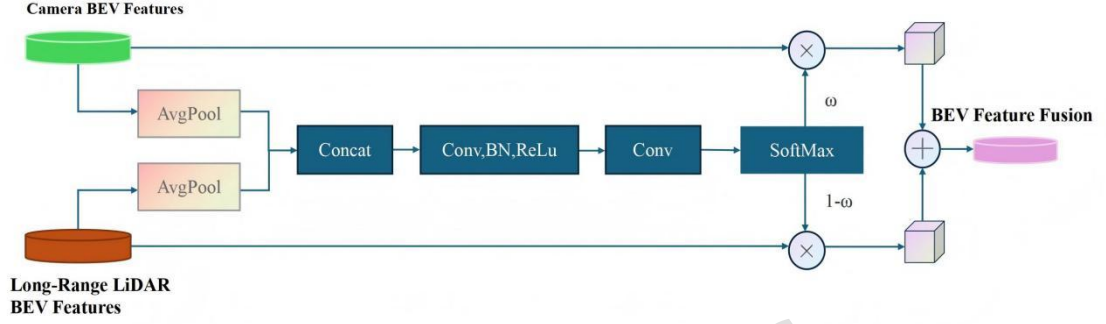


Fig. 5 Structure of the Dynamic Gated Fusion Method

Firstly, for camera and LiDAR features with different resolutions, the camera features C' and LiDAR features Z are unified to the same spatial dimensions through adaptive pooling, addressing the fundamental spatial misalignment issue, as shown in Equation (9).

$$\begin{cases} Z' = ZW_z + b_z \\ C'' = C'W_c + b_c \end{cases} \quad (9)$$

Subsequently, the aligned bimodal features are concatenated along the channel dimension, as described in Equation (10), and then passed through a 1×1 convolution for channel compression to significantly reduce computational complexity. The compressed features are further processed using a 3×3 convolution, as depicted in Equation (11):

$$X_s = \begin{bmatrix} Z'_{11} & \cdots & Z'_{1dz} & C''_{11} & \cdots & C''_{1dc} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ Z'_{B1} & \cdots & Z'_{Bdz} & C''_{Bdc} & \cdots & C''_{Bdc} \end{bmatrix} \quad (10)$$

$$S = \text{Conv}_{3 \times 3}([\text{Conv}_{1 \times 1}(X_s), C_{cam}, C_{lidar}]) \quad (11)$$

In urban road scenarios, the model dynamically adjusts sensor weights based on environmental characteristics: it enhances LiDAR features in areas with clear geometric structures or dense point clouds, while camera features are prioritized in well-lit, texture-rich regions. The Softmax function, by exponential each score, ensures positive values while amplifying the differences. After normalization, the probability distribution that forms the weight map is obtained, as presented in Equation (12).

$$g = \text{Softmax}(s) = \left[\frac{e^{s_1}}{\sum_{j=1}^J e^{s_j}}, \dots, \frac{e^{s_k}}{\sum_{j=1}^J e^{s_j}} \right] \quad (12)$$

The normalization process can be described as follows, where component e^{s_k} applies an

exponential function to each value s_k in the vector to ensure positivity and amplify differences.

$\sum_{j=1}^k e^{s_j}$ sums these exponentiated values, and $\frac{e^{s_1}}{\sum_{j=1}^k e^{s_j}}$ normalizes each value by dividing it by this sum

Finally, the generated weight maps are element-wise multiplied with the original camera and LiDAR features respectively, to achieve position-adaptive feature enhancement and suppression. The weighted features are concatenated along the channel dimension to form a fusion feature that preserves the original bimodal information.

4. Experimental Analysis

4.1 Dataset

Mini NuScenes Dataset

The core purpose of the Mini NuScenes dataset is to serve as an efficient development and debugging tool. It provides a lightweight, low-cost experimental platform by extracting the complete sensor and annotation data from the first 10 scenes of the full dataset.

NuScenes Dataset

The NuScenes dataset considering on the complexity of real-world driving environments is widely applied in the experimental validation³⁵, providing an authoritative testing platform for evaluating model performance. The core purpose of the full NuScenes dataset is to serve as a benchmark for scientific research, encompassing 1,000 scenes and over 300 GB of rich data. It covers a wide range of challenges in different cities, weather conditions, times of day, and driving environments. Each sample is annotated at 2 Hz and includes images from six cameras, covering a 360° horizontal field of view of the ego vehicle.

4.2 Evaluation Indicators

To validate the comprehensive performance of the CRFusion method in semantic map construction tasks, the evaluation is conducted by the **Intersection over Union (IOU)**, **Mean Intersection over Union (mIOU)**, **Average Precision (AP)** and **Mean Average Precision (mAP)** metrics.

Intersection over Union (IOU) measures the degree of overlap between the predicted HD map M_1 and the ground truth HD map M_2 , calculated via the following formula:

$$\text{IOU}(M_1, M_2) = \frac{|M_1 \cap M_2|}{|M_1 \cup M_2|} \quad (13)$$

The similarity between the predicted curve for the single-direction chamfer distance (CD) and the ground truth curve is given by the following Equation (14).

$$\text{CD} = \frac{1}{C_1} \sum_{x \in C_1} \min_{y \in C_2} \|x - y\|_2 \quad (14)$$

where C_1 and C_2 denote the sets of points on the predicted curve and ground truth curve respectively. CD is employed to calculate the spatial distance between the two curves. However, there are the problems by CD alone for HD map evaluation.

Average Precision (AP) measures instance detection capability, where P_r represents the precision at recall rate r .

$$\text{AP} = \frac{1}{10} \sum_{r \in \{0.1, 0.2, \dots, 1.0\}} \text{AP}_r \quad (15)$$

4.3 Metric Optimization Based on the Mini NuScenes dataset

Before the end-to-end evaluation of the CRFusion framework based on the complete Nuscenes data, the Mini Nuscenes dataset is used in advance to verify the feasibility of the three methods

proposed, and the specific verification information is shown in Table 1 and Table 2.

Table 1. IOU detection results on the Mini NuScenes dataset

Method	Modality	Backbone	Epochs	IOU _{ped}	IOU _{Divider}	IOU _{Boundary}	mIOU
HMapNet	C&L	Efficient&PointPillars	30	1.94	0	3.1	1.68
SuperFusion	C&L	ResNet-101&PointPillars	30	2.66	2.6	3.6	2.95
CRFusion	C&L	ResNet-101&PointPillars	30	3.5	1.3	4.5	3.1

Table 2. AP results for the Mini NuScenes dataset

Method	Modality	Backbone	Epochs	AP _{ped}	AP _{Divider}	AP _{Boundary}	mAP
SuperFusion	C&L	ResNet-101&PointPillars	30	1.3	0.4	5.3	2.3
CRFusion	C&L	ResNet-101&PointPillars	30	2.3	1.1	4.7	2.7

The experimental verification in Tables 1 and 2 shows that the IOU_{ped}, IOU_{Boundary}, mIOU, AP_{ped}, AP_{Divider} and mAP of the proposed method work are increased by 0.96 %, 0.9 %, 0.15 %, 1 %, 0.7 % and 0.4 %, respectively, compared with the performances of the Superfusion method. These results preliminary validate the efficacy of the CRFusion framework in mitigating critical challenges such as long-range feature sparsity, incomplete local completion, and insufficient multimodal fusion—in semantic map generation, thereby laying a solid foundation for subsequent large-scale, end-to-end evaluation.

Table 3 presents the experimental results of long-range LiDAR BEV feature prediction for semantic map generation in urban driving environments on the Mini NuScenes dataset.

Table 3. IOU Results at Different Distances for the Mini NuScenes dataset

Method	Modality	0 meter - 30 meter			30 meter - 60 meter			60 meter - 90 meter		
		Divider	Ped	Boundary	Divider	Ped	Boundary	Divider	Ped	Boundary
HMapNet	C&L	5.9	0	3.1	0.6	0.2	0.8	3.5	0	1.2
SuperFusion	C&L	6.8	7.3	5	0.8	0	1	4.9	0	1.5
CRFusion	C&L	3.2	7.2	9.4	1.4	0.4	2.4	3.1	0.085	1.8

Experimental results indicate that the lane boundary detection is improved by 4.4% at 0 meter-30 meter compared with the Superfusion method. At 30 meter-60 meter, the Intersection Over Union (IOU) metrics for lane boundaries, lane dividers, and pedestrian crossings are increased by 0.6%, 0.4%, and 1.4%, respectively. At 60 meter-90 meter, the IOU metrics for lane dividers and pedestrian crossings are increased by 0.085% and 0.6%, respectively. This advantage stems from CRFusion enhancement of feature display through a self-attention mechanism, building upon SuperFusion lidar prediction capabilities.

Table 4 presents the feasibility verification experiments conducted on the Mini NuScenes dataset, aiming to validate the effectiveness and working mechanism of the LRP and DGF methods.

Table 4. Impact of LRP and DGF on the Fusion Performance of the CRFusion Method on the Mini NuScenes dataset

LRP	DGF	Modality	Backbone	IOU _{Divider}	IOU _{Ped}	IOU _{Boundary}	mIOU
×	×	C&L	ResNet-101&PointPillars	0.85	0	1	0.62
×	√	C&L	ResNet-101&PointPillars	1.3	0.03	2.2	1.17
√	×	C&L	ResNet-101&PointPillars	1.2	0.017	2.44	1.21
√	√	C&L	ResNet-101&PointPillars	1.36	0.038	3.4	1.60

The results indicate: (1) When the LRP method is not used and the DGF method is applied, the improvements are 0.45%, 0.03%, and 1.2% 0.55% in lane boundaries, lane dividers, crosswalks, respectively, and mIOU. (2) When the LRP method is used and the DGF method is not applied, the improvements are 0.35%, 0.017%, 1.44%, and 0.59% in lane boundaries, lane dividers, crosswalks, and mIOU. (3) When both the LRP and DGF methods are used, the improvements are 0.51%, 0.038%, 2.4%, and 0.98% in lane boundaries, lane dividers, crosswalks, and mIOU. This advantage stems from CRFusion ability to predict more distant LiDAR BEV features and complete local LiDAR features through LRP, while also achieving superior semantic map construction by adaptively selecting valuable fusion information via the DGF method. The results extremely validate the effectiveness of the CRFusion method.

4.4 Semantic Segmentation Experiments on the Nuscenes Dataset Optimized Based on Metrics

Based on the aforementioned experimental results from the Mini Nuscenes dataset, semantic segmentation experiments by the CRFusion method are conducted on the NuScenes dataset. According to the same experimental settings and data splitting strategy, the proposed CRFusion method is verified compared with several state-of-the-art methods, namely HDMaNet, BEVFusion, and SuperFusion. Tables 5 and 6 present the overall performance of CRFusion and all baseline methods based on the dataset.

Table 5. IOU Results on the NuScenes dataset

Method	Modality	0 meter - 30 meter			30 meter - 60 meter			60 meter - 90 meter			Average IOU		
		Divider	Ped	Boundary	Divider	Ped	Boundary	Divider	Ped	Boundary	Divider	Ped	Boundary
VPN	C	21.1	6.7	20.1	20.9	5.1	20.3	15.9	1.9	14.7	19.4	4.9	18.5
LSS ³⁴	C	35.1	16	33.1	28.5	6.5	26.7	22.2	2.7	20.7	28.9	9.4	27.2
PointPillars ¹²	L	41.5	26.4	53.6	18.4	9.1	25.1	4.4	1.7	6.2	23.7	14.5	30.7
HDMaNet ¹⁴	C&L	44.3	28.9	55.4	26.9	10.4	31.0	18.1	5.3	18.3	30.5	16.6	35.7
BEVFusion ²⁹	C&L	42	27.6	52.4	26.8	11.9	30.3	18.1	3.3	15.9	30.0	16.3	34.2
SuperFusion ³⁰	C&L	47.9	37.4	58.4	35.6	22.8	39.4	29.2	12.2	28.1	38.0	26.2	42.7
CRFusion	C&L	48.26	36.5	59.88	36.8	23.8	41.7	29.3	11.5	29.3	39.6	27.5	44.7

Table 6. AP Results on the NuScenes dataset

Method	Modality	0 meter - 30 meter			30 meter - 60 meter			60 meter - 90 meter			Average AP		
		Divider	Ped	Boundary	Divider	Ped	Boundary	Divider	Ped	Boundary	Divider	Ped	Boundary
VPN	C	16.2	3.4	30.5	17.1	4.1	30.2	13.3	1.5	21.1	15.6	3.1	27.5
LSS	C	24.0	9.9	39.3	23.9	5.7	38.1	19.2	2.2	26.2	22.5	6.2	34.8
PointPillars	L	24.6	18.7	49.3	15.9	7.8	36.8	4.1	1.9	9.2	15.6	10.1	32.7
HDMaNet	C&L	30.5	20	54.5	23.7	4.2	26.4	15.2	4.2	26.4	23.6	11.7	43.1
BEVFusion	C&L	25.8	19.1	47.6	20.3	4.0	18.5	12.5	4.0	18.5	20	11.6	35.4
Superfusion	C&L	33.2	22.5	58.0	25.1	10.7	26.9	24.1	10.7	38.2	24.7	19.2	50.1
CRFusion	C&L	35.2	21.0	59.5	26.3	11.2	27.8	24.0	9.5	39.1	26.2	20.5	51.9

Experimental results demonstrate that: (1) The IOUs of the lane boundary and lane separation line from 60 meter to 90 meter are increased by 0.1 % and 1.2 %, respectively, and the AP of the lane boundary is increased by 0.9 %, indicating that the results on CRFusion is significantly better than

those on the Superfusion method. It is because that features are enhanced by the self-attention mechanism, and valuable information is selected by the dynamic gated fusion. (2) The proposed CRFusion method achieves corresponding metric improvements compared to Superfusion in the distance ranges of 0 meter-30 meter, 30 meter-60 meter, and 60 meter-90 meter. The advantage arises because the other three methods (HDMaNet, BEVFusion, and Superfusion) rely on direct channel concatenation and convolution for multi-modal feature fusion, which overlook inter-channel interactions and adopt overly simplistic fusion strategies.

Table 7 and Table 8 show the experimental results for high-definition map construction and BEV map segmentation tasks on different fusion methods, respectively.

Table 7. Comparison of IOU under Different Fusion Methods on the NuScenes dataset

Method	Modality	Backbone	Epochs	IOU _{Divider}	IOU _{ped}	IOU _{Boundary}	mIOU
Fusion	C&L	ResNet-101&PointPillars	30	38.0	26.2	42.7	35.6
Dynamic Gated Fusion	C&L	ResNet-101&PointPillars	30	38.6	26.0	43.1	35.9

Table 8. Comparison of AP under Different Fusion Methods on the NuScenes dataset

Method	Modality	Backbone	Epochs	AP _{Divider}	AP _{ped}	AP _{Boundary}	mAP
Fusion	C&L	ResNet-101&PointPillars	30	24.7	19.2	50.1	31.3
Dynamic Gated Fusion	C&L	ResNet-101&PointPillars	30	25.2	18.7	51.2	31.7

The experimental results show that the mIOU is increased by 0.3 %, and the mAP is increased by 0.4 %. It shows that the DGF method plays a vital role in the BEV feature fusion, and can automatically select valuable information from the camera BEV feature and the LiDAR BEV feature, so as to better perform feature fusion.

4.5 Ablation Experiment

To evaluate the performance of the LRP and DGF methods within the CRFusion algorithm, ablation experiments are conducted on the NuScenes dataset. Four experimental setups are designed as follows: (1) We train the model without either the LRP or DGF method. (2) We train the model with the DGF method. (3) We train the model with the LRP method. (4) We train the model with both the LRP and DGF methods. The corresponding experimental results are presented in Tables 9 and 10. Herein, LRP is the method for the LiDAR long-range feature prediction, while DGF is used to assign weights to fuse LiDAR and camera BEV features by the gate in the different environments.

Table 9. Impact of Different Methods on the IOU of CRFusion on the NuScenes dataset

LRP	DGF	Modality	Backbone	IOU _{Divider}	IOU _{ped}	IOU _{Boundary}	mIOU
×	×	C&L	ResNet-101&PointPillars	38.0	26.2	42.7	35.6
×	√	C&L	ResNet-101&PointPillars	38.6	26.0	43.1	35.9
√	×	C&L	ResNet-101&PointPillars	38.5	26.5	43.6	36.2
√	√	C&L	ResNet-101&PointPillars	39.6	27.5	44.7	37.3

Table 10. Impact of Different Methods on the AP of CRFusion on the NuScenes dataset

LRP	DGF	Modality	Backbone	AP _{ped}	AP _{Divider}	AP _{Boundary}	mAP
×	×	C&L	ResNet-101&PointPillars	24.7	19.2	50.1	31.3
×	√	C&L	ResNet-101&PointPillars	25.2	18.7	51.2	31.7
√	×	C&L	ResNet-101&PointPillars	25.5	19.1	51.5	32.0

√	√	C&L	ResNet-101&PointPillars	26.2	20.5	51.9	32.9
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As shown in Tables 9 and 10, the IOU and AP for lane boundaries, lane dividers, and crosswalks are improved on the full Nuscenes dataset when LRP for long-range LiDAR BEV feature prediction are performed. Due to the DGF, LiDAR and camera BEV features are effectively leveraged in the fusion process, so mIOU and mAP receive a higher promotion compared to the metrics in Superfusion. Specifically, the IOU for lane boundaries, lane dividers, and crosswalks are improved by 1.6%, 1.3%, and 2%, respectively. The AP are improved by 1.5%, 1.3%, and 1.8%, respectively. The mIOU and mAP are increased by 1.7% and 1.6%, respectively. The improvements significantly enhance the network's fusion performance, confirming that long-range prediction of LiDAR features and dynamic gated fusion effectively improve semantic segmentation performance in complex environments.

4.6 Semantic Map Visualization

To visually demonstrate the effectiveness of the CRFusion method, the red vehicle indicates the car's current position. Each map spans 90 meters relative to the vehicle's length. Different colors denote distinct instances of high-definition map elements. For ground-truth high-definition maps, green red and blue represent lane boundaries, lane cent relines and pedestrian crossings, respectively. The visual comparative maps of the Real Semantics BEVFusion, and SuperFusion are presented as Fig. 6.

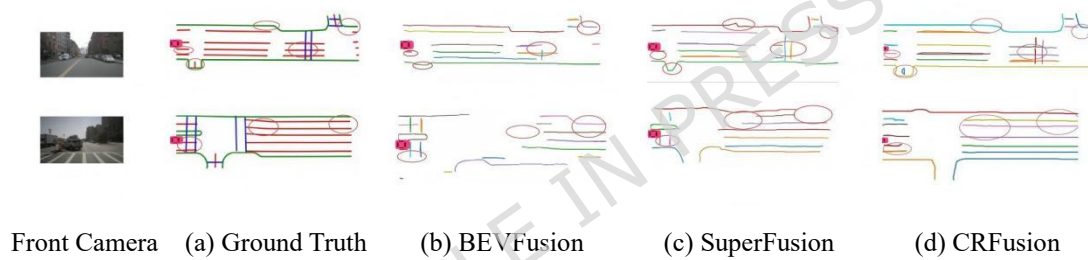


Fig. 6 Visualisation of the Improved Semantic Map

A comparison of the first-row images reveals that our method produces superior structural continuity in distant areas (marked by red circles), such as lane boundaries and crosswalks, effectively mitigating the fragmentation and loss of detail observed with Superfusion. The result validates the efficacy of our LRP module in long-range feature completion. As shown in the second-row images (red circles), the details of elements such as vehicles and lane markings more clearly and completely on CRFusion method are reconstructed. It indicates that the DGF fusion mechanism enhances the quality of the semantic map.

Overall, comparative analysis with Superfusion demonstrates improvements in the structural integrity and detail reconstruction of map elements achieved by our method, and CRFusion effectively enhances the quality of semantic map generation is validated.

5. Conclusion

To address the challenges of sparse long-range point clouds, LiDAR-camera modal feature misalignment, and poor adaptability to complex environments in multimodal BEV semantic mapping, the CRFusion fusion framework was proposed. After the preliminary experiments based on Mini NuScenes dataset, the experiments on the full NuScenes dataset confirmed its effectiveness, where the mIoU for lane boundaries, lane dividers, and crosswalks reached representing gains of 1.6%, 1.3%, and 2.0% over the baseline Superfusion method. Ablation results indicated that each component of the CRFusion framework contributed significantly to the overall performance improvement. Moreover,

visualizations showed that the semantic maps generated by CRFusion exhibited smoother lane-line continuity and clearer long-range crosswalk contours. Collectively, these quantitative and qualitative results validated the effectiveness, robustness, and superiority of CRFusion in the semantic map generation task, offering a more reliable perception solution for autonomous driving in complex urban environments.

Data availability

The data provided in this study can be obtained from the corresponding author Y.G.

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Author contributions

Y.G. and T.W. designed the research. Y.G., T.W., Q.C., X.Y. and Y.Y. wrote and edited the paper. Y.G. and T.W. conceived the experiments, T.W. conducted the experiments, Y.G. and T.W. analyzed the results. Q.C. and F.J. check the paper to eliminate errors.

All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

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